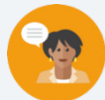


4.3 Identifying Key Features of Lines From Tables and Equations

Lesson Introduction

 Benchmark	MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.		 Learning Targets	- I can Identify key features of a linear function from a graph. - I can identify key features of a linear function given a table or equation.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.		MA.912.F.1.7 Compare key features of two functions each represented algebraically, graphically, in tables or written descriptions.	
 Essential Questions	- Which key features can be determined from each representation, graph, equation, or a table of values? - Which forms of a line are best for interpreting key features?			
 Lesson Narrative	Lesson 3 extends learning from Lesson 2 to include interpreting key features from a table or equation. While Lesson 2 positioned students to explore the key features from a graph, Lesson 3 provides an opportunity for Guided Instruction to encourage students to make connections between what they discovered from the graphs and what they interpret from tables and equations.			
 Component Summary	4.3.1 Warm-Up (~5 min.)	Students fill in missing values to create equivalent equations of lines. <i>(SE p. 175)</i>	4.3.4 Your Turn! (15-20 min.)	Students engage in the analysis and critique of three student responses for applying a variety of strategies in identifying slope and y-intercept from an equation. <i>(SE p. 177)</i>
	4.3.2 Collaboration (5-15 min.)	Students complete a collaboration on key features from a table and a graph. <i>(SE p. 175)</i>		
		4.3.3 Guided Instruction (15-20 min.)	Students connect the graph, key features, and multiple representations of the same line. <i>(SE p. 176)</i>	4.3.5 Wrap-Up (~5 min.)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Domain - Range - Intercepts		(Optional) Foldables or Frayer Model for organizing key features	
			Additional Preparation	
			(Optional) Prepare a model or example for students to follow using a Frayer Model	

 Teacher Tips	Common Misconceptions • Students may struggle with a conceptual understanding of domain or range being all real numbers if a line does not “look” like it extends to all real numbers. • Students with gaps in solving equations or number sense may struggle to make the connections between the different forms of a line. • Students may not initially recognize the x-intercept and y-intercept from a table.	Things to Consider • Consider using Grade 8 Unit 7 Lesson 3 for additional scaffolding when interacting with point-slope and slope-intercept form. • Consider using On-Ramp videos on supporting skills like determining and interpreting the slope and y-intercept, which may be useful for small group rotations in this lesson.
	Literacy and Language Routines Think-Pair-Share 4.3.2 Collaboration is designed to be a collaborative learning experience. Since key features from a table have not yet been explicitly taught, a think-pair-share strategy would be beneficial for this activity. Allow students to grapple with the problem first and then discuss with their partner to build confidence and refine their responses.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection 4.3.4 Your Turn! provides students with three different student justifications for finding the slope and y-intercept of the same equation. Students will analyze each explanation to determine its mathematical validity and explain why it is or is not an accurate explanation.
	 Differentiate Instruction Continue to monitor students for differentiating between skills – students who may be proficient with identifying domain and range may struggle with number sense in converting between forms of a line, or vice versa. Those nuances will not be fully captured in the Wrap-Up unless explicitly asked. Use each activity as an opportunity to make observations that will inform the possible small groups in the following lesson. The next lesson is the final lesson in a three-part lesson series on key features. 4.3.5 Wrap-Up for this lesson is a self-reflection confidence scale. Consider pairing this with observations from 4.3.4 Your Turn! to determine misconceptions and gaps to inform learning in part three of the following lesson. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

4.3.1 Warm-Up: Equations of Lines

Component Overview: This activity offers a creative entry point for students to access equivalence using a variety of forms for the equation of a line. While forms would not be formally discussed in this activity, some students might begin making those connections for you to build on throughout the lesson.



4.3.1 Warm-Up: Equations of Lines

- Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$y = -\frac{\boxed{}}{\boxed{}}x + \boxed{} \text{ is the same line as}$$

$$\boxed{}x + \boxed{}y = \boxed{}$$

$$y = -\frac{3}{1}x + 4 \text{ and } 6x + 2y = 8$$

or

$$y = -\frac{4}{1}x + 3 \text{ and } 8x + 2y = 6$$

or

$$y = -\frac{9}{3}x + 4 \text{ and } 6x + 2y = 8$$



Consider what scaffolding could look like if students are stuck or need additional entry points. Provide the values for the first equation for students to fill in the second, or vice versa.



For a challenge, have students create a problem similar to this one that other students would be able to solve with the same conditions.

4.3.2 Collaboration: Key Features of a Linear Function From a Table

Component Overview: Students collaborate to determine which key features can be determined from a table of values.



4.3.2 Collaboration: Key Features of a Linear Function from a Table

1. A table of values that represents a function is shown.

x	y
-2	-9
-1	-6.5
0	-4
1	-1.5
2	1
3	3.5

- a. With your partner, discuss which key features of this function, if any, can be determined from the table of values.

- ☐ Domain
- ☐ Range
- ☒ Slope
- ☒ Increasing/Decreasing
- ☐ x -intercept
- ☒ y -intercept



This collaboration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group Guided Instruction. Some students may be able to determine the x -intercept from the table of values. Additional scaffolding questions are provided, if needed, but this component should be primarily student directed.

- Why does a table of values not provide enough information to determine the domain and range?
- Can we determine the x -intercept from this table of values? Can it be determined from any table? Why or why not?

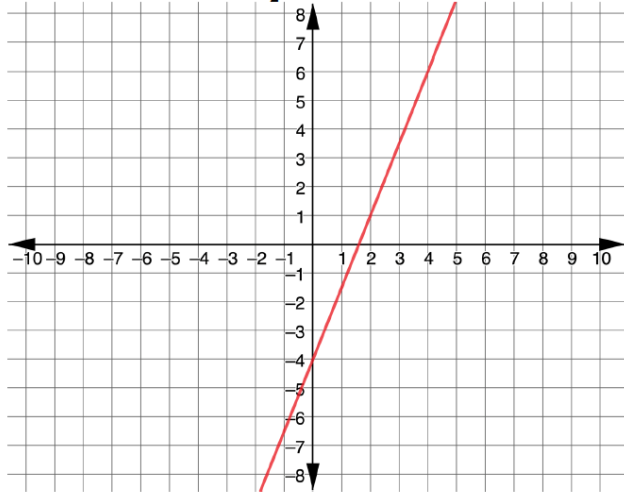


Think-Pair-Share

Utilize an instructional routine to position students to discover and develop strong conjectures that lead to mathematical conclusions. Give students 1 minute to determine the key features before turning to their partner to share findings. Students can be asked to why some key features could not be found. This process of critical thinking, constructing mathematical arguments, and analyzing the responses from others builds students' confidence and abilities related to MA.K12.MTR.4.1.

4.3.2 Collaboration: Key Features of a Linear Function From a Table (continued)

- b. Use the table of values to graph the function it represents, $f(x) = \frac{5}{2}x - 4$.



- c. Find the key features of the function $f(x) = \frac{5}{2}x - 4$.

Domain	Range	As x -values increase, y -values ...
<i>All real numbers</i>	<i>All real numbers</i>	<i>Increase</i>
Slope	x -intercept	y -intercept
$\frac{5}{2}$	$1\frac{3}{5}$	-4



You can use this activity to determine potential differentiation on student graphing abilities.

For kinesthetic learners, consider projecting a visual of the coordinate grid available on display. Have students come up individually to plot the points from the table of values and have a final student connect all the points to make the line.



As students work on questions 1b-c, prompt them with the following:

- Why are the domain and range all real numbers?
- Would we know this with only a table of values?
- How do we know the line continues/ keeps going?

4.3.3 Guided Instruction: Different Forms of a Line

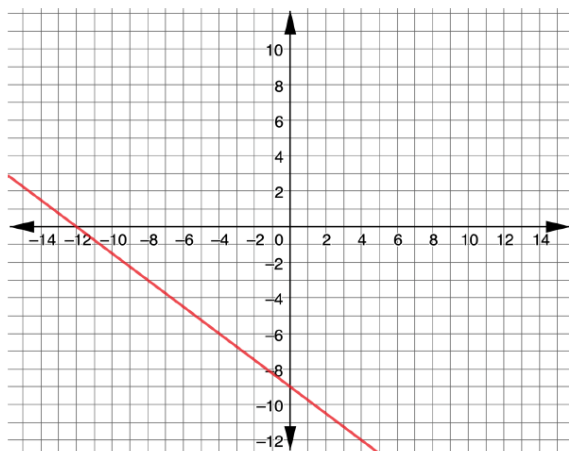
Component Overview: Students will continue identifying key features through the process of graphing the equation of a line and then identifying those key features.



4.3.3 Guided Instruction: Different Forms of a Line

1. Consider the equation $y + 3 = -\frac{3}{4}(x + 8)$.

a. Graph $y + 3 = -\frac{3}{4}(x + 8)$.



b. Find the key features of $y + 3 = -\frac{3}{4}(x + 8)$.

Domain	Range	As x -values increase, y -values ...
<i>All real numbers</i>	<i>All real numbers</i>	<i>Decrease</i>
Slope	x -intercept	y -intercept
$-\frac{3}{4}$	-12	-9



Monitor math vocabulary for Parts A-B. Consider using talk moves like, “Can you restate...” or “Can you reword...” leveraging math vocabulary with common phrases like:

- “Where it crosses...” with “The y -intercept of...”
- “The x and y ...” with “The domain and range...”
- “As x gets bigger....” With “As x -values increase...”



During instruction for questions 1a-b, prompt students with the following:

- What form of line is the given equation?
- What key feature is easy to determine from the equation?



Try a foldable for key features as scaffolding for students who best learn by creating and producing diagrams like Frayer Models or interactive foldables.

4.3.3 Guided Instruction: Different Forms of a Line (continued)

2. The equation $y + 3 = -\frac{3}{4}(x + 8)$ is just one way of writing an equation or a function that represents the line you graphed. The table shows the four common representations of the same line.

Point-Slope Form	Slope-Intercept Form
$y + 3 = -\frac{3}{4}(x + 8)$	$y = -\frac{3}{4}x - 9$
Standard Form	Function Notation
$3x + 4y = -36$	$f(x) = -\frac{3}{4}x - 9$

- a. Which forms of the line listed are the easiest to find the slope?

Answers will vary. The slope-intercept form is easiest because the value of m is the same as the slope. The point-slope form is easiest because the value outside of the parentheses is always the slope.

- b. Which forms of the line listed are the easiest to find the y -intercept?

Answers will vary. The slope-intercept form is easiest because I know the value of b is always the y -intercept. The standard form is easiest because I can plug a 0 in for the x -value and solve a basic one-step equation to find the y -intercept.



Students engaged with formal definitions of each form in Unit 3. For additional scaffolding, consider providing Math Nation Glossary Videos for each form, or the lesson videos from Unit 3 for students to use at home.



Consider having students share their answers to questions 2a-b. Have students determine if they agree or disagree with each other and try to have groups convince others to join them in their thinking. Consider adding a kinesthetic element to the learning by making it a Four Corners activity.



Consider prompting students with the following:

- Which forms are more challenging for you to work with than others? Why or why not?

4.3.4 Your Turn!

Component Overview: Students analyze and critique the reasoning of other students in this activity, in which only one of three students do not find the correct slope. Provide an opportunity for students to independently analyze, draw conclusions, and mathematically prove their findings.



4.3.4 Your Turn!

1. A group of students are discussing how to find the slope and y -intercept of the line with the equation $9x - 15y = -75$.

- a. Use the description each student gives to complete their work.

Malcolm	Olivia	William
To find the slope and y -intercept, I rewrite the equation to slope-intercept form.	To find slope, I mentally move the x -term to the other side and divide by the coefficient of the y -term. To find the y -intercept, I divide the value the equation is equal to by the coefficient of the y -term.	To find slope, I write the coefficient of the x -term divided by the coefficient of the y -term. To find the y -intercept, I write the value the equation is equal to divided by the coefficient of the y -term.
Malcolm's Work	Olivia's Work	William's Work
$9x - 15y = -75$ $-15y = -9x - 75$ $\frac{-15}{-15}y$ $= \frac{-9}{-15}x - \frac{75}{-15}$ $y = \frac{9}{15}x + 5$ or $y = \frac{3}{5}x + 5$	Slope = $\frac{-9}{-15}$ or $\frac{3}{5}$ y -Intercept = $\frac{-75}{-15}$ or 5	Slope = $\frac{9}{-15}$ or $-\frac{3}{5}$ y -intercept = $\frac{-75}{-15}$ or 5

- b. Which student would NOT find the correct slope and/or y -intercept?

William would not get the correct slope and y -intercept.



Students who finish early can create an explanation for how to find the x -intercept of the same equation. Have them share with each other and determine if they are correct.

Additionally, ask students to give advice to the incorrect student on why they made a mistake and how they would fix it.



Consider prompting students with the following:

- When doing an analysis of someone else's work, were you able to look and tell what went wrong?
- What process did you use to find the errors?



In question 1b, William's misconception occurs because of using a shortcut or formula for $Ax + By = C$, rather than fluency of students using different forms.

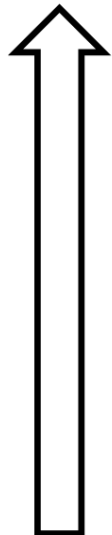
4.3.5 Wrap-Up: Reflection

Component Overview: Students will complete a confidence scale, rating their level of confidence in identifying key features of a linear function. This activity should be done independently to gain an accurate self-reflection from each student on their confidence with the content of this lesson.



4.3.5 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding of identifying key features of a linear function.



I'm so confident, I can explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and listened, but I am finding this challenging. I will make sure I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

Answers will vary. Students should color the arrow to the level of their understanding of the topic.



Students are self-reflecting on their confidence on this lesson. Compare students' classroom performance with their confidence level to design differentiated experiences.

For example, what is provided for a student who has high confidence and low accuracy would be different than low confidence and high accuracy.

If students need additional support, give them an example problem to analyze to gauge their confidence.